

Citing code borrowed from other sources

Although writing code may seem different than writing papers, the same standards of acknowledgment apply. If you rely on someone else's program, you *must* credit that person.

```
513 class Set {
514     private Object elements[];
515     private int MAX = 500;
516     private int pos;
517
518     public Set() {
519         elements = new Object[MAX];
520     }
521     public void reset() {
522         for (int i = 0; i < MAX; i++) {
523             elements[i] = null;
524         }
525     }
526     public boolean addElement(Object x) {
527         if (containsElement(x))
528             return false;
529         int i;
530         i = 0;
531         while (i < MAX) {
532             if (elements[i] == null) {
533                 elements[i] = x;
534                 return true;
535             }
536             i++;
537         }
538         return false;
539     }
540     public boolean containsElement(Object x) {
541         int i;
542         i = 0;
543         while (i < MAX) {
544             if (elements[i] != null
545                 && elements[i].equals(x)) {
546                 return true;
547             }
548             i++;
549         }
550         return false;
551     }
552     public Object findElement(Object x) {
553         for (int i = 0; i < MAX; i++) {
554             if (elements[i] != null
555                 && elements[i].equals(x)) {
```

To cite borrowed code there is no need to copy the original. Instead, you should add a comment in your code that makes two things clear:

1. The code you are using is not your own original work; and
2. the location of the code you borrowed, usually with a URL.

For example, suppose you are writing a JavaScript program. You need a function that sorts an HTML table by columns, and you find the following code online at the site <http://example.com/sort.js>:

```
function sortTableArray(tableToSort) {
    // The code from the original function
}
```

If you borrow this function exactly in your own code, with no changes (except perhaps changing the function name), then you should cite it like this:

```
/*
 * Code copied from function sortTableArray, found at:
 * http://example.com/sort.js
 */
function sortTable(tableToSort) {
    // The code from the original function
}
```

If you borrow this function with some modifications (whether minor or major) in your own code, then you should cite it like this:

```
/*
 * Code modified from function sortTableArray(), found at:
 * http://example.com/sort.js
 */
function sortTable(tableToSort) {
    // The code from the original function
}
```

Notice that the only difference between the two ways citations is that the exact copy says “code copied from...”, and the modified version says “code modified from ...” In either case, you have acknowledged that you are not the original author, and you reference the source so that others can locate it.

Important: If you don't include proper in-text citations, then you commit plagiarism.

Citing a computer program in a written text

In-text citations

If you cite a computer program in the body of your text, then you must include a reference to it in the text. Here is an example:

My research examines a number of different systems. *Raster3D* (Merritt & Bacon, 1997) is an example of a suite of graphics programs for UNIX systems.

Important: If you don't include proper in-text citations, then you commit plagiarism.

The reference list

If you cite a computer program in your text, then you must include a full reference to it in a reference list at the end of your assignment (see the example below). The reference list provides the information that is needed by a reader who wishes to find the code that you have cited.

Merritt, E.A. and Bacon, D.J. (1997) *Raster3D Photorealistic Molecular Graphics* (Version 2.0) [Computer program]. Available at <http://skuld.bmsc.washington.edu/raster3d/html/raster3d.html> (Accessed 19 August, 2015)

The reference for a program should include:

1. Author (if given) – can be individual(s) or corporation
2. Date (if given)
3. Title of program (in italics and capitalized)
4. Version (in round brackets)
5. Form e.g. Computer program (in square brackets)
6. Availability: URL and date accessed if downloaded from the internet.

Note: scholars from different disciplines have different expectations about citation and referencing style. For example, scientists use a different style compared to lawyers. The examples above use the *Harvard* format, but we recommend that you ask your teachers which style they expect you to use. A number of common styles are listed below, with links to related style guides:

- [APA](#) (commonly used in Social Sciences)
- [CSE](#) (commonly used in Natural and Applied Sciences)
- [Chicago](#) (commonly used in Natural and Social Sciences)
- [IEEE](#) (commonly used in Engineering)
- [MLA](#) (commonly used in Humanities)
- [Harvard](#) (commonly used in Social Sciences)

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